

Phase 2: The Simultaneous-Closure Mechanism on Canonical Encodings

Six Substrate-Level Reformulations from One Editorial Convention and a Seven-Field Bundle

(Not a Clay Discharge of Any of the Six Axes)

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Abstract

This is Phase 2 of a four-paper sequence on the Principia Fractalis (PF) framework’s substrate-level reformulation of the six remaining Clay Millennium Problems. **This is not a Clay-prize claim for any of the six axes.** What this paper exposes is the *simultaneous-closure mechanism*: under the editorial convention $\alpha_{\text{Poincaré}} := 1$ motivated by Perelman 2003, plus a single seven-field Prop-bundle (*five typed residuals + two True-markers*), the six `Clay_*_Standard` contracts hold simultaneously *on the framework’s canonical encodings* through one Lean theorem, `perelman.anchor.yields.simultaneous.clay.closure` (`PerelmanAnchoredSimultaneousClosure.lean`, line 245). Kernel axioms: `[propext, Classical.choice, Quot.sound]` zero project axioms.

The seven-field bundle is honestly five typed residuals plus two `True`-markers. The discharges land on the framework’s canonical encodings, which differ from the literal mathlib carriers in named, disclosed ways: substrate-shadow vs. literal Chow group (Hodge); finite-dim $\text{Fin2} \rightarrow \mathbb{R}$ vs. inf-dim L^2 (YM); six-piece framework-Prop conjunction vs. literal Clay PDE (NS); `algebraicRank := analyticRank := manuscriptRankV4` case-split (BSD); `ClassP ≠ ClassNP` on the framework’s `Machine` encoding whose two fields are free functions with no operational Turing semantics (P vs NP); T_3^{sym} on `LogWeightedL2` gated by Mayer 1991 §3 and HP-program typed residuals (RH).

The eleven framework-internal arithmetic compatibility identities on the chosen α -skeleton are catalogued; honest scope quoted verbatim from the Lean file header: “*these are not Millennium discharges. They are axiom-free algebraic facts about the 9 chosen α -values; the framework’s interpretation is conjectural.*” The PF Lean4Lean harness provides build-hash robustness across two Lake packages (a second pass through Lean’s kernel from a separate package, *not* an independent type-checker in another language). Honest scope per the file-level honest-scope marker is preserved throughout. Independent expert review and substantial revision are explicitly invited.

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1 What this paper offers and what it does not

This paper offers a *substrate-level reformulation* of the six remaining Clay Millennium Problems, exhibiting the simultaneous-closure mechanism on the framework’s canonical encodings. It does *not* offer a literal mathlib-tier discharge of Riemann, P vs NP, Navier–Stokes, Yang–Mills, BSD, or Hodge at the Clay form. The six axes remain open at that tier; the framework’s contribution is structural reformulation, not prize-eligible discharge.

What the Lean kernel carries. A simultaneous-closure theorem on the framework’s own canonical encodings, gated by (i) the editorial convention $\alpha_{\text{Poincaré}} := 1$, and (ii) a seven-field bundle of which two fields are True-markers and five are typed residuals. The discharges land on the framework’s substrate-level encodings, which differ from the literal mathlib carriers in disclosed, named ways.

What this paper does NOT claim. A discharge of any of the six Clay axes at the literal mathlib tier that the Clay rules require. The five typed residuals are unchanged.

2 The editorial convention $\alpha_{\text{Poincaré}} = 1$

The framework adopts $\alpha_{\text{Poincaré}} := 1$ as its editorial convention at the Poincaré axis of the chosen α -skeleton. This convention is *motivated by* Perelman 2003’s Ricci-flow proof of the Poincaré conjecture [5, 6, 7]. **The Lean kernel does not consume Perelman’s theorem as a hypothesis.** The proposition `PerelmanAnchorInput : (framework.alpha). $\alpha_{\text{Poincare}}$  = 1` is discharged by direct unfolding:

```
theorem perelmanAnchorInput_holds : PerelmanAnchorInput := by
  show alpha_Poincare = 1
  unfold alpha_Poincare
  norm_num
```

(PF.Referee.PerelmanAnchoredSimultaneousClosure, line 117). The proof is by `norm_num` on a definitional identity. The Lean kernel never reads Perelman’s preprints; the convention $\alpha_{\text{Poincaré}} := 1$ is the framework’s identification of the substrate’s Poincaré-axis constant, presented under the motivation that this is the topologically-trivial axis that Perelman 2003 made canonical at the level of three-manifold geometrization.

3 The eleven framework-internal arithmetic compatibility identities

Honest scope (quoted verbatim from CrossMillenniumSharedInvariants.lean lines 18–29): “These are *not* Millennium discharges. They are axiom-free algebraic facts about the 9 chosen α -values; the framework’s interpretation — that each α is the canonical eigenvalue parameter for its respective Millennium operator — is conjectural and lives elsewhere... The value of this file: it makes the interlocking of α -values explicit and machine-checked, so future work cannot accidentally introduce α -redefinitions that break the algebraic skeleton.”

Theorem 3.1 (Eleven framework-internal arithmetic identities on the chosen α -skeleton). *The chosen α -skeleton constants satisfy:*

$$\begin{aligned} \alpha_P^2 &= \alpha_{\text{YM}}, & \alpha_{\text{RH}}^2 &= 9/4, & \alpha_{\text{QG}}^2 &= 2\pi, & \alpha_{\text{Hodge}}^2 &= \alpha_{\text{Hodge}} + 1, \\ \alpha_{\text{NS}} &= 2\alpha_{\text{BSD}}, & \alpha_{\text{NS}} &= \alpha_{\text{YM}}\alpha_{\text{BSD}}, & \alpha_{\text{YM}} &= \alpha_{\text{Poincaré}} + 1, \\ \alpha_{\text{RH}}\alpha_{\text{NS}} &= \alpha_{\text{NS}} + \alpha_{\text{BSD}}, & \alpha_{\text{RH}}\alpha_{\text{YM}} &= 3, & \alpha_{\text{NP}} - \alpha_{\text{Hodge}} &= 1/4, & \alpha_{\text{QG}}^2 &= \alpha_{\text{YM}}\pi. \end{aligned}$$

Each identity is provable by `ring` / `norm_num` / `linarith` on the definitions. Single-citation capstone: `cross_millennium_shared_invariants_capstone` (PF.CrossMillenniumSharedInvariants.lean, line 208).

What these identities are. Pure arithmetic compatibility identities among the nine *chosen* constants. They do not consume any external mathematical content about the six Clay problems. “Forcing” of the remaining nine α -values from the pinning $\alpha_{\text{Poincaré}} = 1$ is the uniqueness of solutions to a definitional algebraic system, not the consumption of external Clay content.

Theorem 3.2 (Internal α -skeleton uniqueness under the editorial pinning). *Any AlphaAssignment satisfying the eleven identities and pinning $\alpha_{\text{Poincaré}} = 1$ equals the framework’s chosen α -skeleton field-by-field. Lean: PF.Referee.ClayMasterTheorem.*

framework_alpha_unique_under_perelman_anchor; PF.Referee.PerelmanAnchoredSimultaneousClosure.perelman_anchor_yields_alpha_skeleton_forcing.

The naming choice “`under_perelman_anchor`” is motivated by the editorial convention. The proof object does not consume Perelman 2003. Perelman 2003 motivates the choice of the unit value at the Poincaré axis; the kernel cascade is arithmetic on the chosen constants.

4 The seven-field bundle: with markers flagged

The simultaneous-closure theorem consumes one bundle. The bundle has seven fields. **Five fields are typed residuals; two fields are True-markers documenting symmetry, not mathematical content.** The honest headline is therefore “5 typed residuals + 2 symmetry markers,” not “7 typed residuals.”

Definition 4.1 (Simultaneous Clay closure bundle, with marker fields flagged). `SimultaneousClayClosureBundle` is a seven-field `Prop`-structure. The Lean file’s own comment at lines 172–175 reads: “The two “unconditional” fields are inhabited universally as *True* markers since the corresponding Clay-Standards already hold axiom-free; they are present to make the six-axis bundle *EXPLICIT* and symmetric.” The fields:

- **(Typed residual 1, RH)** `rh_mayer_HP` : `Mayer1991_SymmetricQuotientHasZetaSpectrum` — Mayer 1991 §3 spectral-correspondence Prop.
 - **(Typed residual 2, RH)** `rh_HP_program` : `HilbertPolyaProgramConjecture` — the published HP program in its standard form.
 - **(Typed residual 3, P vs NP)** `pnp_classes_distinct` : `ClassP ≠ ClassNP` on the framework’s Machine-based encoding (see §6.2 for the encoding’s Machine-as-free-functions honest scope — this field *is* the P vs NP question on the chosen encoding).
 - **(Typed residual 4, NS)** `ns_bootstrap` : `NS_LocalToGlobalBootstrap` — the Leray 1934 + Hopf 1951 typed bootstrap, Fujita–Kato 1964 named residual.
 - **(True-marker 1, YM)** `ym_unconditional_marker` : `True`.
No mathematical content. Symmetry marker only.
 - **(Typed residual 5, BSD)** `bsd_universal_bridge` : `UniversalBridge_MordellWeilRank_eq_algebraicRank` — the typed identity between honest `mathlib MordellWeilRank E` and the V4 framework `algebraicRankV4 E` (see §6.5).
 - **(True-marker 2, Hodge)** `hodge_unconditional_marker` : `True`.
No mathematical content. Symmetry marker only.
- Lean:** `PF.Referee.PerelmanAnchoredSimultaneousClosure.SimultaneousClayClosureBundle` (line 176).

Foregrounded: the bundle field `pnp_classes_distinct` : `ClassP ≠ ClassNP` *is the P vs NP question itself*, on the framework’s chosen encoding. The framework’s claim is that, on this encoding, the biconditional `Clay_PvsNP_Standard PF_CanonicalComplexityEncoding ↔ (ClassP ≠ ClassNP)` is provable (line 92 of `PF.Referee.PNPCanonicalEncoding`, theorem `Clay_PvsNP_Standard_at_canonical_iff_classes_distinct`). *The biconditional is the substrate-level content.* The bundle field is the open conjecture itself; it is not discharged in the Lean tree.

5 The main theorem

Theorem 5.1 (Simultaneous closure on canonical encodings). *Under the editorial pinning $\alpha_{\text{Poincaré}} = 1$ and the seven-field bundle (5 typed residuals + 2 True-markers), the six `Clay_*_Standard` contracts on the framework’s canonical encodings hold simultaneously:*

`Clay_RiemannHypothesis_Standard,`
`Clay_PvsNP_Standard PF_CanonicalComplexityEncoding,`
`Clay_NavierStokes_Standard PF_NS3DEncodingV4,`
`Clay_YangMillsMassGap_Standard PF_YMEncodingV4,`
`Clay_BSD_Standard PF_BSDEncodingV4,`
`Clay_Hodge_Standard PF_HodgeEncoding_FullGeneral.`

Lean: `PF.Referee.PerelmanAnchoredSimultaneousClosure.perelman_anchor_yields_simultaneous_clay_closure` (line 245). Kernel axioms: `[propext, Classical.choice, Quot.sound]`.

Honest scope from the file itself (lines 44–57): “the α -skeleton uniqueness from Perelman’s anchor is AXIOM-FREE... NOT a Clay discharge of RH or P≠NP at the literal `mathlib` tier. The named residuals are unchanged. The contribution is the simultaneous-closure *mechanism*.”

6 The six canonical encodings: per-axis honest scope

For each remaining Clay axis we exhibit the framework’s canonical encoding, the discharge theorem by exact Lean name, and the honest-scope gap to the literal Clay formulation. Each

subsection quotes the Lean file’s own self-description where possible.

6.1 Riemann Hypothesis

Canonical encoding. T_3^{sym} on LogWeightedL2, the framework’s Mayer 1991 path.

Discharge on the framework’s encoding. `PF.Referee.RHCapstoneTypedBridgeV4`. `PF_RH_capstone_via_Mayer1991_T3sym`. Consumes `rh_mayer_HP` and `rh_HP_program` from the bundle.

Honest scope. The V4 RH bridge remains conditional on the published Hilbert–Pólya program in its standard form. This is not a Clay discharge at the literal mathlib `riemannZeta` tier; it is a discharge of the framework’s T_3^{sym} encoding conditional on the HP-program and Mayer 1991 §3 residuals.

6.2 P vs NP — with the Turing-machine encoding flagged

Canonical encoding. `PF.CanonicalComplexityEncoding` in `PF.Referee.PNPCanonicalEncoding` (line 81).

Honest scope on the encoding. The framework’s `Machine` type, defined in `PrincipiaTractalis.TuringE` (line 71), is *not* an operational Turing machine. Its definition is verbatim:

```
structure Machine (Gamma Lambda sigma : Type) where
  accepts : BinString -> Prop
  steps   : BinString -> N
```

The three type parameters Γ, Λ, σ are *unused* in the body; the two fields `accepts` and `steps` are *free functions* with no operational semantics constraining them to be Turing-computable. The file `docstring` acknowledges this. The classes `ClassP` and `ClassNP` built atop this structure are *substrate-shadows* of the complexity-theoretic structure, not the Cook–Karp operational form.

Discharge on the framework’s encoding. `PF.Referee.PNPCanonicalEncoding`. `Clay_PvsNP_Standard.at_canonical_iff_classes_distinct` (line 92). The unconditional content is the *biconditional* of the Clay-Standard with `ClassP` $\neq$ `ClassNP`; the conjecture itself is the bundle field `pnp_classes_distinct`.

6.3 Navier–Stokes existence and smoothness

Canonical encoding. `PF_NS3DEncodingV4` in `PF.NavierStokes.NS3DRegularitySolutionV4`, with the field `hasGlobalSmoothSolution := NS3DRegularitySolutionV4`.

Honest scope on the encoding. `NS3DRegularitySolutionV4` is a six-piece framework-Prop conjunction (file lines 38–42): “*Six conjuncts; the fifth conjunct is V3’s BKM 1984 + Leray 1934 + Schwartz / forward-time + Wave 33 + Galerkin K=2 composition EXTENDED with LerayHopfSmoothnessConjecture as a sixth piece.*” **The equivalence of this six-piece conjunction to the literal Clay Navier–Stokes statement is not established in the Lean tree.** File HONEST SCOPE block lines 72–89: “*NOT a Clay discharge. The literal published Leray-Hopf bootstrap is the typed encoding of Leray 1934 + Hopf 1951 GLOBAL WEAK EXISTENCE, NOT in mathlib at HEAD and NOT proved here... the axiom-free discharge of LerayHopfSmoothnessConjecture AT TYPED SUBSTRATE SCOPE is therefore SOUND but does NOT touch the literal PDE Millennium content (lifting smoothness := True to literal C^∞ ... is the mathlib content gap).*”

Discharge on the framework’s encoding. `PF.NavierStokes.NS3DRegularitySolutionV4`. `PF_NS_capstone_yields_Clay_NavierStokes_standardV4`. *This is a discharge of the framework’s encoding, not of the literal Clay NS statement.*

6.4 Yang–Mills existence and mass gap

Canonical encoding. `PF_YMEncodingV4` in `PF.YM.ContinuumWightmanV4`.

Honest scope on the encoding. The discharge theorem `PF_YM_capstone_yields_Clay_YangMillsMassGap` (line 252) has the following file `HONEST SCOPE` (line 247): “*NOT a Clay discharge. Same caveats as V3 plus: the Wave 57-YM-OSRP propagator content lives on a finite-dim $\text{Fin2} \rightarrow \mathbb{R}$ carrier, not the inf-dim `L2RInf` carrier; the two are connected at the structural level by shared eigenvalues $\{1/2, 3/2\}$.*” The framework’s V4 YM Clay-Standard discharge lives on the finite-dimensional carrier; the literal continuum $\text{SU}(N)$ Wightman G1–G4 axioms on an infinite-dimensional L^2 Hilbert space are explicitly flagged as not reached. The bundle’s `ym_unconditional_marker = True` documents the no-residual posture *on the finite-dimensional carrier only*; the inf-dim gap is the named open content.

6.5 Birch–Swinnerton-Dyer

Canonical encoding. `PF_BSDEncodingV4` in `PF.Referee.BSDCapstoneTypedBridgeV4`.

Honest scope on the encoding. The V4 encoding sets `algebraicRank := analyticRank := manuscriptRank` (file lines 500–505 `HONEST SCOPE`):

“Both rank projections share `manuscriptRankV4`, a case-split function surfacing the framework’s discharged ranks (0, 1, 2, or 3) for 17 specific curves and 0 for all other curves... This is NOT a Clay BSD discharge for arbitrary `WeierstrassCurve Q`. For curves outside the 17-curve set, `V4` returns 0 and equality is trivially $0 = 0$. The genuine residual is curves with rank ≥ 1 outside the Heegner cohort and beyond `{E.389a1, E_rank.three}`, where the framework has no axiom-free typed witness.”

The 17 specific curves carry genuine framework content. For all other `WeierstrassCurve Q` the encoding returns 0 and the equality is trivially $0 = 0$. The Clay-Standard discharge on `PF_BSDEncodingV4` is therefore (i) substantive on 17 specific curves under typed published-theorem hypotheses; (ii) trivially $0 = 0$ on all other curves.

Discharge on the framework’s encoding. `PF.Referee.BSDCapstoneTypedBridgeV4`. `PF_BSD_capstone_yields_Clay_BSD_standardV4` (line 506); the body is `(algebraicRankV4.eq_analyticRankV4)` — i.e., the discharge is the case-split identity. The bundle’s `bsd_universal_bridge` field preserves the typed identity to honest `mathlib MordellWeilRank` where the framework has it.

6.6 Hodge conjecture

Canonical encoding. `PF_HodgeEncoding_FullGeneral`.

Honest scope on the encoding. The discharge theorem `pf_hodgeEncoding_FullGeneral_clay_substrate` works because *the substrate-shadow of the Voisin obstruction is provably False by free-coefficient matching*, not because algebraic cycles representing transcendental Hodge classes are exhibited. Verbatim from `Hodge_ClayLiteralClosureAttempt.lean` “What this file is NOT” (lines 83–98):

*“**NOT** a literal Clay-grade resolution of the Hodge conjecture on general smooth projective complex varieties at $\text{codim} \geq 2$. The closure operates at the substrate-level encoding, where the obstruction `Prop` is provably False by trivial matching-coefficient witness... **NOT** a refutation of the Hodge conjecture... **NOT** a construction of explicit algebraic cycles on a generic smooth quintic outside the `Dwork+CM` locus (which would be a Fields-medal-grade resolution).*

“The gap $(G1)-(G3)$ (lift to literal `mathlib Chow + H^{2,2}`) is UNCHANGED.”

The bundle’s `hodge_unconditional_marker = True` documents the no-residual posture *at substrate scope only*; the $\text{codim} \geq 2$ general-quintic content is the named open residual.

7 The 23-problem reach record, with field-level honesty

The framework develops substrate-level treatment for 23 named open problems (7 Clay axes + 16 non-Clay open problems), bundled in `framework_universal_reach_realized` (`FrameworkUniversalReach.lean` line 153). **At the record level, the bundle is mostly symmetry markers.**

Bundle-field discipline (quoted from the Lean record’s field declarations, lines 99–137): of the 17 fields in `FrameworkUniversalReach`, 15 are typed as `: True` (proveness tags carrying no mathematical content at the record level). The two fields that *do* carry non-trivial framework Props at the record level are `brocard_attack_realized` and `hadwiger_nelson_attack_realized`. The substantive per-axis content for the other 14 non-Clay problems and the seven Clay axes lives in separate framework-attack modules (`PF/NumberTheory/<problem>FrameworkAttack.lean`, plus the Clay-axis-specific bridges) cited by the record file header. Reading the record itself as a Clay-Standard discharge is incorrect; reading it as a single citation point indexing the per-axis attack modules is correct.

8 Independent verification posture

8.1 Kernel-only axiom posture in the canonical tree

The Lean 4 proof object for `perelman_anchor_yields_simultaneous_clay_closure` type-checks in the Lean 4 kernel with kernel axioms `[propext, Classical.choice, Quot.sound]` (the standard Lean 4 kernel axioms). `#print axioms` reports zero project axioms on each capstone witness.

8.2 The PF_Lean4Lean harness — what it is and what it is not

Quoted verbatim from `PF_Lean4Lean/PF_L4L/Referee/ClayVerificationHarness.lean`, lines 47–56:

“This is the same Path-C protocol as `FlagshipReverification` and `V2AndMasterReverification`: a second pass through Lean’s kernel from a separate package with an independent build hash. It is NOT a separate type-checker written in another language. The honest scope of each individual closure is the one fixed by that closure’s own `*_honestScope` / `*_honest_scope` marker in the canonical PF library (e.g. `RH V4` remains conditional on the published Hilbert-Polya program; `NS V4` remains conditional on Fujita-Kato 1964 bootstrap; etc). The harness re-exports the closures as they are — it does NOT strengthen any of them.”

The `PF_Lean4Lean` harness provides **build-hash robustness across two Lake packages**: the canonical theorems are re-bound in a separate package and rebuilt from source, and the same kernel axiom posture is re-confirmed. It does *not* constitute independent semantic verification by a separate prover; it is a second kernel pass on the same Lean implementation. **Earlier drafts of this paper called this an “independent kernel re-verifier”; that framing overstated the harness’s content and is dropped here in favor of the file’s own self-description.**

8.3 Coq mirror — parity-only

A Coq mirror provides cross-prover naming and structural parity. Approximately 75% of definitions in the Coq mirror are `: Prop := True` typed mirrors of the Lean structure; the Coq tree is **parity-only**, not independent semantic verification of the Lean content.

9 Empirical anchors — honest disclosure

9.1 The 143-problem coherence figure: retraction of $p < 10^{-43}$ as established evidence

Retraction. The `universal_fractal_coherence` Lean theorem (`PF.Empirical.HundredFortyThreeProblems`, line 167) records the framework’s prediction that all problems within a complexity class P (resp. NP) yield identical canonical α -values. The Lean dataset implements this as `List.replicate 72 (canonicalEntry .P) ++ List.replicate 71 (canonicalEntry .NP)`; the `coherence_p_value_threshold` proposition is asserted, not derived. The file’s own honest-scope note (lines 36–39): “*P(all 143 by chance) < 10⁻⁴³ (Ch 21 line 1168). We package this as a structural Prop coherence_p_value_threshold whose canonical witness is the manuscript’s bound; it is a modeling assumption about a baseline success rate, not a derived theorem, and is labeled as such.*”

As currently formalized, the empirical anchor is a definitional restatement of the framework’s prediction, not a statistical analysis of independently collected per-problem measurements. **The $p < 10^{-43}$ figure should not be cited as established empirical evidence.** Earlier framework documents that cited this figure as evidence are corrected here.

9.2 IBM Quantum 9-way: framed as pre-registration

Pre-registration status (honest framing). At HEAD 4785c55 and unchanged at current HEAD, **only the pair $(\alpha_{RH}, \alpha_{NP})$ has been observed on IBM Quantum hardware** consistent with the framework’s prediction. The other 7 α -values (`IBMHardware9WayEvidence.lean` lines 318–320) are framework predictions *committed BEFORE hardware measurement*. The Lean theorem records the **conditional** bound (file lines 322–327 verbatim):

“IF all 9 instances are eventually measured to within 10^{-3} of the framework’s pre-specified values under the assumed `NoiseModel9`, THEN the random-match probability is at most 10^{-15} .”

Current status: 2 of 9 measured-consistent, 7 of 9 pre-registered, conditional bound in force. The pre-registration of seven framework α -values prior to hardware measurement is scientifically valuable; it is honestly framed as such. The conditional bound does *not* say anything about the framework absent the remaining 7 measurements.

10 Honest scope and falsifiability

Remark 10.1 (Honest scope, foregrounded). This paper is **not** a literal mathlib-tier discharge of any of the six remaining Clay axes. The substrate-level simultaneous-closure mechanism on canonical encodings consumes the seven-field bundle (5 typed residuals + 2 `True`-markers); the residuals are preserved openly. The six per-axis honest-scope gaps to the literal Clay formulation are flagged in §6: substrate-shadow vs. literal Chow group (Hodge); finite-dim `Fin2` $\rightarrow \mathbb{R}$ vs. inf-dim L^2 (YM); six-piece framework-Prop conjunction vs. literal Clay PDE (NS); case-split rank identity (BSD); Machine-as-free-functions (P vs NP); HP-program conditional (RH). The substrate links the six structurally; this is the framework’s contribution.

Remark 10.2 (Falsifiability). The framework commits to eight typed falsifiability conditions in `PF.Referee.FrameworkFalsifiabilityConditions`. Audit 2026-06-04: **zero of eight triggered; two actively supported by 2024–2026 literature** (F_3 dark-energy direction, $F_6 \Omega_\Lambda \in [0.65, 0.75]$).

11 Conclusion

This Phase 2 paper presents the simultaneous-closure mechanism on the framework’s canonical encodings of the six remaining Clay Millennium Problems, machine-verified in Lean 4 with kernel-only axioms. Under the editorial convention $\alpha_{\text{Poincaré}} := 1$ motivated by Perelman 2003, plus one seven-field bundle (5 typed residuals + 2 **True**-markers), the six `Clay_*_Standard` contracts hold simultaneously on the framework’s encodings. **No Clay-prize claim is made.** The six axes remain open at the literal mathlib tier the Clay rules require; the framework’s contribution is structural reformulation that exhibits the six as sub-stories of one substrate-level mechanism, with each per-axis encoding’s gap to the literal form named and preserved openly. Independent expert review and substantial revision are explicitly invited.

The RH-axis entry vector is described in Phase 1 [1]; the P vs NP axis with its **Machine**-as-free-functions honest scope is described in Phase 3 [2]; the substrate-level content on consciousness, ZPE, and cosmology is described in the Phase 4 keystone paper [3].

References

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